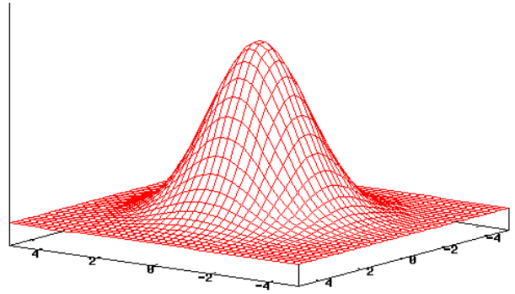
Exercise #2

Function Approximation MLP

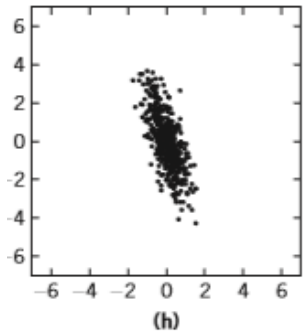
The Data



- Given a 2-D Gaussian Function

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}[det(\mathbf{C}_{\mathbf{X}})]^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})'\mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})\right)$$

$$\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix}\mu_1\\ \mu_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad \mathbf{C}_{\mathbf{x}} = \begin{bmatrix}0.3 & -0.5\\ -0.5 & 2\end{bmatrix}$$



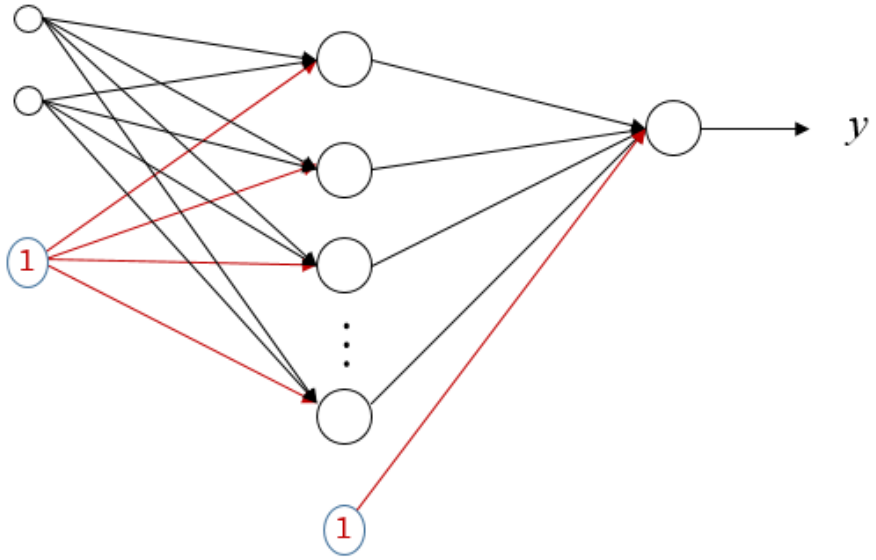
- Generate A training samples $Y_i = (x_{i1}, x_{i2}, y_i)$ randomly

$$-2 \le x_{i1} \le 2, \quad -4 \le x_{i2} \le 4, \quad \text{and } 0 \le y_i \le 1$$

- Generate 100 test samples similarly

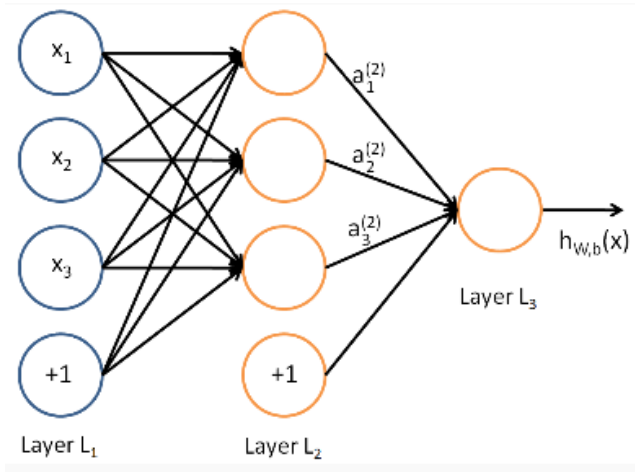
Network Structure

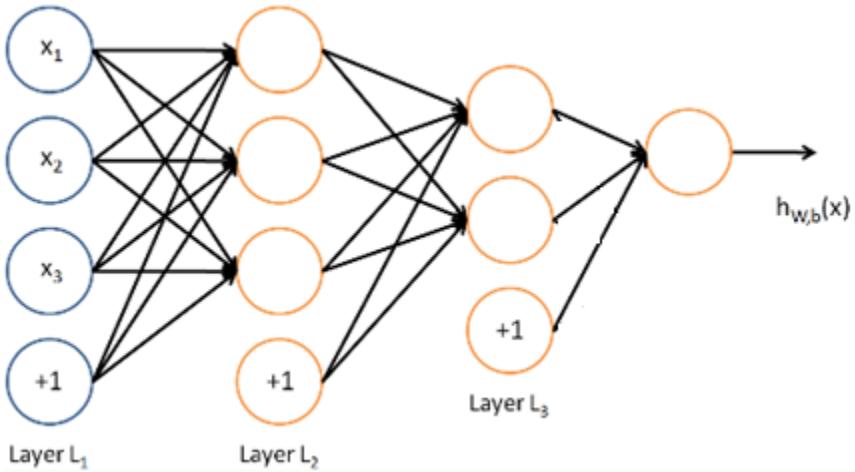
- MLP with one hidden layer
  - 2 input nodes (+ bias)
  - 1 output node
  - B hidden nodes



Network Structure

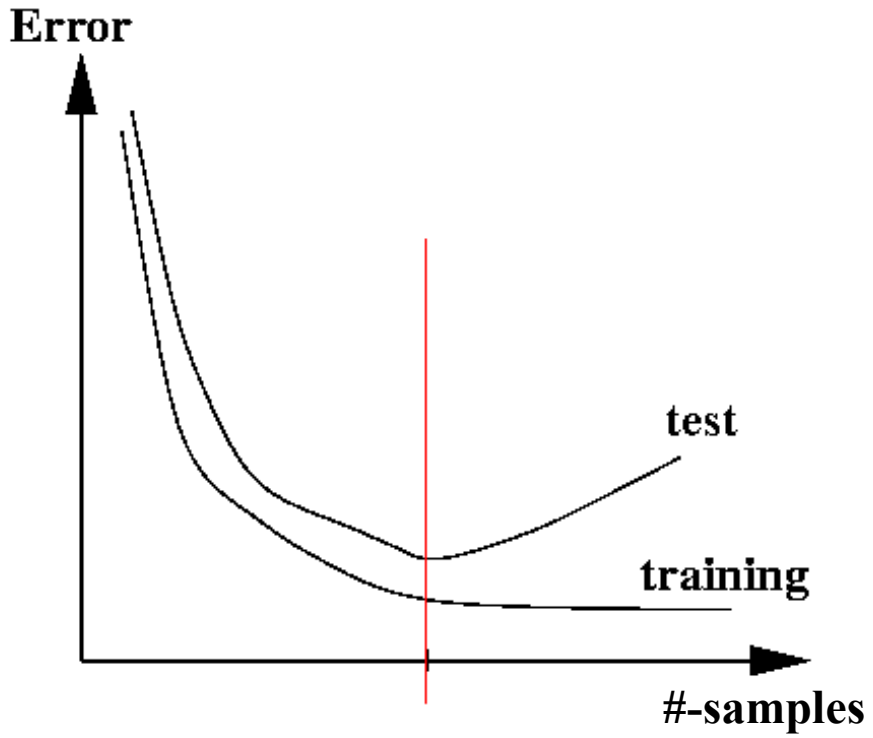
- MLP with more than two hidden layers
  - 2 input nodes (+ bias)
  - 1 output node
  - C hidden layers





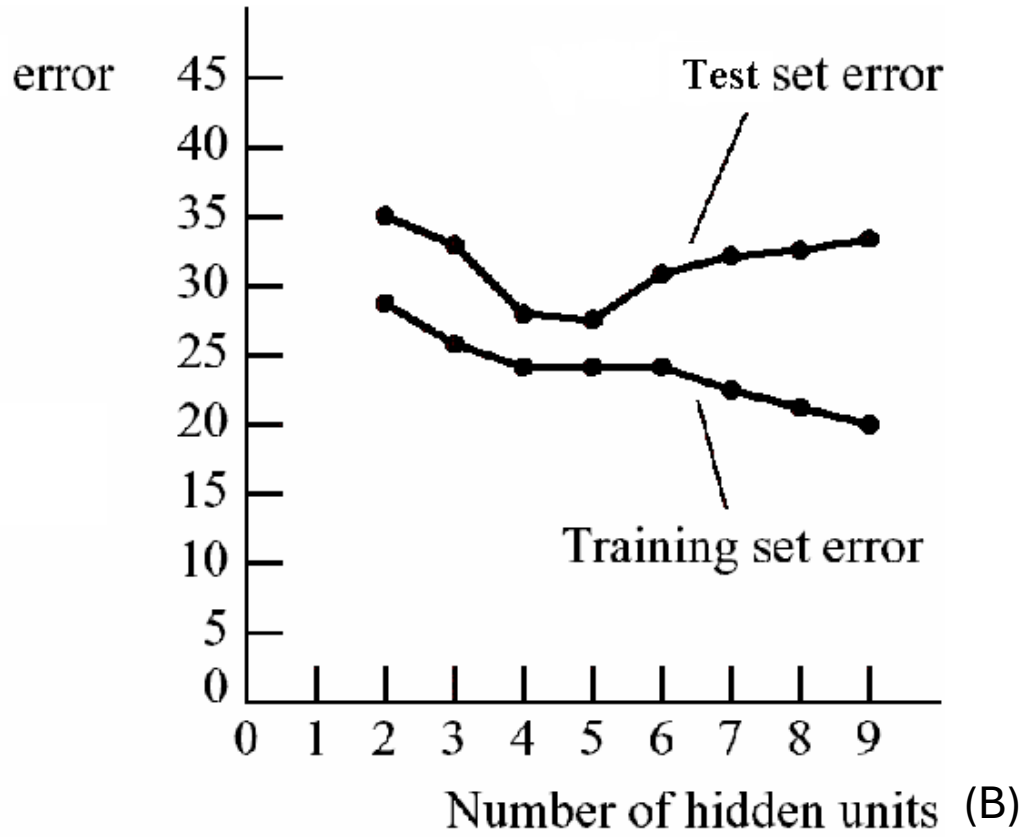
Report (1/3)

- Performance against the # training samples (A)



Report (2/3)

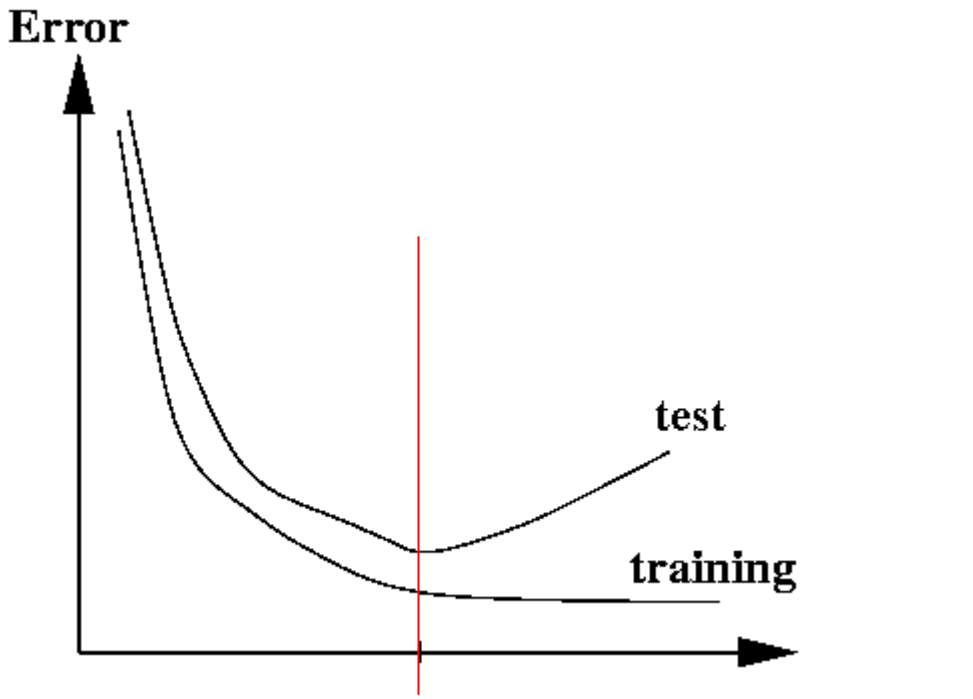
- Performance against the # hidden nodes (B)



(B)

Report (3/3)

- Performance against the # hidden layers (C)



Pay Attention

- Submission
  - Report file
    - Brief summary for source codes
    - Results for many experiments with various parameter settings
    - Analysis and explanation about the results in your own way
  - Source Code using Pytorch

- Due
  - Try to use sample program shown in the class
  - Source code & results will be examined in the class

THANK YOU!